

ATLAS COORDINATE SYSTEM

The ATLAS detectors' coordinate system allows scientists to accurately and consistently describe how particles travel through the detector.

Z-AXIS

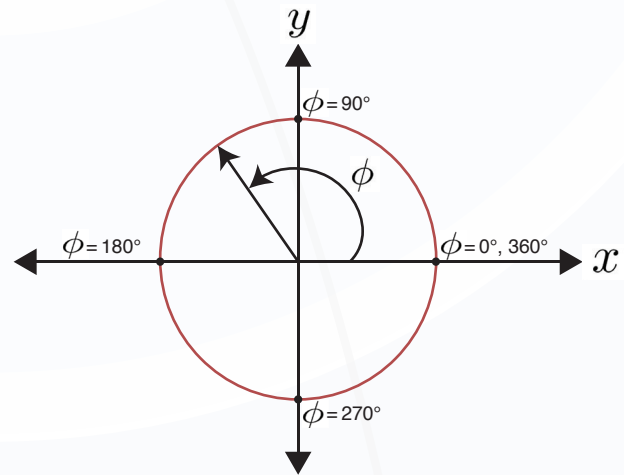
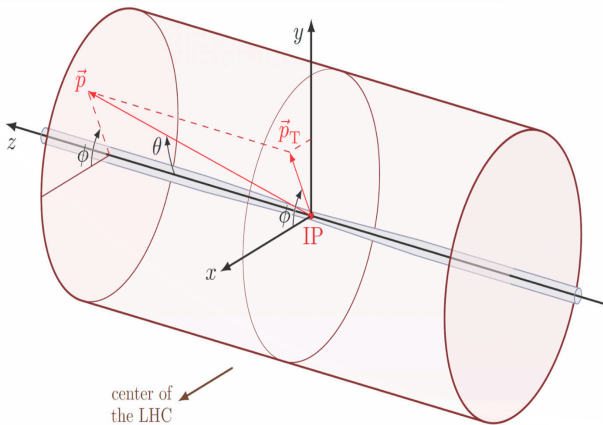
The Large Hadron Collider (LHC) accelerates particles at very high speeds along a path called the beam line. The z-axis runs along the beam line. Particle collisions occur at the origin (or interaction point (IP)), which is chosen to be at the center of the ATLAS detector. The +z axis points in the direction of the beam that is circulating counterclockwise in the accelerator.

THETA (θ)

Theta (θ) is the polar angle and it represents the angle between a particle's momentum vector and the z-axis. It ranges from 0° to 180° , where $\theta = 0^\circ$ corresponds to motion directly along the +z direction and $\theta = 180^\circ$ corresponds to motion along the -z direction. Theta only describes how far a particle's path is from the z-axis, not where it is located around the z-axis. Due to the detector's cylindrical symmetry, rotations around the z-axis will leave θ unchanged.

PHI (Φ)

The x-y plane is perpendicular (transverse) to the z-axis and it represents a cross section (or slice) of the detector. The +x-axis points towards the center of the LHC and +y points up. Phi (Φ) is the azimuth angle and it describes where the particle's path is located around the detector. It ranges from 0° to 360° . Assuming $\theta = 90^\circ$, $\Phi = 0^\circ$ corresponds to motion directly along the +x direction and $\Phi = 90^\circ$ corresponds to motion directly along the +y axis.



PSEUDORAPIDITY (η)

The pseudorapidity (η), pronounced "eta," describes how far a particle is tilted from the x-y plane. Particle physicists prefer using η instead of θ because differences in pseudorapidity between particles from the same collision are independent of the observer's motion along the z-axis. This is shown in the figure: the cones have different widths in theta ($\Delta\theta$) but share the same width in eta ($\Delta\eta$). The relationship between eta and theta is given by:

$$\eta \equiv -\ln \left[\tan \left(\frac{\theta}{2} \right) \right]$$

